

Ideological discrepancy between publishers and news content is linked with audience engagement and consensus on Facebook

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Abstract

Political news on social media rarely circulates in isolation: audiences actively engage, react, and clash. Whether these interactions reflect agreement or conflict may depend on the ideological discrepancy between publishers and the news content they share. This study investigates this relationship using Facebook posts linking to political news during a Brazilian presidential election. We analyze five dimensions of engagement: ideological discrepancy between publishers and content, emotional responses, audience consensus, toxicity in posts, and content topics. Our results show that ideological discrepancy is associated with differences in engagement, exhibiting a nonlinear pattern: consensus declines under conditions of very high ideological mismatch and, in our data, also under very high alignment, while toxicity increases primarily under extreme mismatch. A statistical model indicates that emotional valence, toxicity, and ideological discrepancy are the factors most strongly associated with consensus. Among highly partisan publishers, higher toxicity is associated with increased audience consensus, suggesting that hostile discourse may co-occur with in-group agreement in strongly ideological contexts. Overall, these findings highlight how ideological discrepancy, emotional reactions, and interaction dynamics are associated with consensus and polarization in online political engagement.

1 Introduction

The rise of social media as a primary source of political information has significantly transformed news consumption and how individuals engage with political content [1]. Platforms like Facebook employ engagement-driven algorithms that have been shown to amplify selective exposure and reinforce ideological divides [2, 3]. By curating personalized content based on users' past interactions, these platforms may contribute to the formation of "filter bubbles," informational ecosystems in which exposure to divergent views is limited and pre-existing beliefs are reinforced [4, 5]. This process has been associated with increased polarization, as homophilous networks [6] promote the formation of ideologically homogeneous groups where misinformation can spread quickly [7].

These structural conditions carry important affective consequences. Because echo chambers amplify political beliefs while limiting exposure to opposing viewpoints [4, 8], hostility between opposing groups may intensify [9], and interactions may become increasingly toxic [10]. Such dynamics are particularly salient in electoral contexts, where social media plays a central role in voter mobilization and narrative formation. The 2018 Brazilian presidential election provides a relevant example: Jair Bolsonaro's

campaign extensively used digital platforms to disseminate political narratives around corruption and public security [11], producing network dynamics characterized by tight clustering around ideological poles [12]. These dynamics reflected an intensification of affective polarization, more pronounced regarding candidates than parties, and accompanied the growth and radicalization of the right-wing [13]. This polarization laid the groundwork for subsequent electoral cycles, with analyses indicating that polarization continued to increase through the 2022 election [14].

Despite the breadth of this literature, most existing work focuses on network exposure [15, 16], content diffusion [7, 8], or structural polarization metrics [17, 18]. Comparatively, less attention has been given to how ideological discrepancy between publishers and the news content they share relates to the affective quality of participatory engagement. Addressing this gap, the present study examines how the ideological discrepancy between *publishers* (public Facebook pages and verified accounts that share news links) and the news content they share relates to the engagement of the *audience* (the broader community of Facebook users whose reactions to those posts constitute our engagement signals), using Brazil’s presidential election as a case study. Two research questions guide this inquiry: (RQ1) How does ideological discrepancy between publishers and shared news content relate to audience engagement? (RQ2) Which factors are most strongly associated with variations in audience consensus?

To address these questions, we draw on a dataset of Facebook posts sharing political news links from Brazil’s 2018 presidential election. We characterize ideological discrepancy using the Bias Discrepancy (Δ_b), the absolute difference between each publisher’s estimated political bias (b_p) and the political leaning of the news content they share (b_c). Low Δ_b indicates ideological alignment, whereas high Δ_b reflects ideological mismatch. We then examine its relationship with three dimensions of audience engagement: the Reaction Score (RS), which captures the emotional valence of reactions; the Consensus Index (CI), which measures the degree of agreement in reaction polarity among users; and post Toxicity (Tox), which quantifies the presence of hostile language in the publisher’s text. News content is further characterized through zero-shot topic classification. Together, these constructs operationalize RQ1. To address RQ2, we model CI using a Generalized Linear Mixed Model (GLMM) with a Beta distribution, estimating how ideological discrepancy, emotional valence, toxicity, and content topics relate to variations in audience consensus while accounting for differences across publishers and shared news items.

This study makes two contributions to the literature. First, building on prior work that has focused primarily on exposure, diffusion, or sentiment, we introduce a reaction-centered analytical framework that integrates measures of ideological discrepancy, affective response, and audience consensus to examine how polarization is enacted through participatory interaction on Facebook, rather than inferred solely from network structure. Applying this framework to a large dataset of Facebook posts, we identify an asymmetry in polarized engagement during the 2018 Brazilian presidential election: posts shared by right-leaning publishers tend to exhibit higher levels of audience consensus and more positive reactions, whereas interactions around posts from left-leaning publishers appear comparatively more fragmented and toxic — though these patterns should be interpreted in light of the relative underrepresentation of right-leaning publishers in the dataset.

Second, we contribute to understanding toxicity in online news consumption by showing that its relationship with audience consensus exhibits a nonlinear pattern and is context-dependent. Toxicity escalates primarily under conditions of extreme Bias Discrepancy. Among highly partisan publishers, elevated toxicity was associated with higher rather than lower consensus, suggesting that hostile language may, in some contexts, foster in-group cohesion. Together, these findings move beyond broad claims

of algorithmic determinism often associated with the Filter Bubble perspective [5], instead highlighting how specific interaction dynamics are associated with cohesion or fragmentation in online political engagement.

The remainder of this paper is structured as follows: Section 2 reviews related work. Section 3 presents the problem statement. Section 4 describes the dataset and methodology, Section 5 presents empirical results, and Section 6 concludes the study.

2 Related work

Political polarization in online environments has become a central research topic in computational social science. A growing body of work investigates how user behavior, social ties, and algorithmic systems contribute to ideological echo chambers [4, 19] and political homophily [6, 20, 21], with findings spanning multiple platforms and methodologies. These dynamics continue to evolve. Social media platforms increasingly shape perceptions of polarization [22], while social sorting reinforces affective polarization across digital interactions [23]. Network analyses further reveal nuanced ways in which partisanship can bridge or exacerbate online echo chambers [24]. Against this background, three interrelated lines of research inform the present study: (i) how algorithmic exposure shapes the formation of echo chambers, (ii) how existing polarization metrics capture ideological divisions in online networks, and (iii) how affective responses and emotional reactions structure participatory engagement with political content.

Research on algorithmic curation highlights how recommendation systems shape users' exposure to ideologically diverse content. For instance, [15] found that individual choices, more than algorithms, limit cross-cutting content on Facebook, while [16] showed that algorithmic amplification of low-credibility sources can exacerbate polarization during elections. Similar dynamics appear on YouTube, where personalization has been associated with the reinforcement of echo chambers [25], and where cross-cutting exposure can paradoxically increase polarization [9]. Subsequent work [1] demonstrated that these effects vary across political contexts, with highly polarized environments showing stronger resistance to cross-cutting information. These dynamics were also observed in Brazil's 2018 election [12], where network analyses revealed tight clustering around ideological poles. Recent investigations of cross-partisan interactions further complicate this picture. While [26] observed significant engagement among users with differing political views on Twitter, these interactions were often characterized by negative, nonconstructive stances, suggesting that breaking echo chambers does not necessarily lead to constructive dialogue. Similarly, examining downstream effects on Reddit, [27] showed that receiving a cross-party reply rarely increases subsequent engagement with out-party communities, instead reinforcing in-party participation. Collectively, these findings suggest that studying exposure patterns alone is insufficient to fully understand how polarization manifests in online interactions.

A second line of research focuses on measuring polarization in online networks. Metrics such as $RP(H)$ [28] and controversy frameworks [17, 29] have been used to quantify ideological divisions in retweet networks. These approaches provide valuable insights into the structural organization of polarized communication, particularly in diffusion-based platforms such as Twitter. However, they primarily capture patterns of information flow and network structure, making them less suited to measuring engagement dynamics on platforms like Facebook, where interaction often occurs through reactions and comments rather than content resharing.

A third line of research highlights the role of affective dynamics in online political communication. Prior work shows that textual features play a central role in shaping

affective polarization, revealing how sentiment, emotion, and misinformation structure political discourse online. Political alignment can often be inferred directly from language use and framing [30]. Moreover, false or misleading news tends to spread more rapidly when it evokes high-arousal emotions such as anger [8]. Platform-specific emotional dynamics further influence engagement patterns. For example, sadness has been shown to play a prominent role in the sharing of political content on Facebook [2], reinforcing belief alignment within ideologically homogeneous networks [7]. These affective asymmetries may also be amplified by algorithmic curation mechanisms that promote emotionally engaging content [31]. Together, this body of work highlights the importance of emotional responses in structuring political interaction online. On platforms such as Facebook, these emotional responses are expressed directly through reaction mechanisms, which provide observable signals of collective audience behavior.

Against this backdrop, the present study shifts the analytical focus from diffusion and network exposure toward participatory engagement. Specifically, we examine how reactions, a primary mode of interaction on Facebook, enact and reveal affective polarization during Brazil’s 2018 presidential election. Rather than inferring polarization solely from network structure or diffusion patterns, we operationalize it through reaction-based behavioral signals that capture how audiences emotionally respond to politically aligned or misaligned content. In doing so, we investigate a dimension of online political interaction that existing polarization metrics and platform-comparative studies have only partially addressed.

3 Problem Statement

The objective of this paper is to analyze user engagement with political news. In particular, we aim to characterize how users behave when exposed to content from different sides of the political spectrum. More formally, we consider a set of N_c political news items, where each item c is associated with a political bias $b_c \in [-1, +1]$, ranging from extreme left to extreme right.

Each item c may be posted by one or more publishers, and each publisher p is associated with a political bias score $b_p \in [-1, +1]$, indicating the side of the political spectrum with which they are aligned. In addition, each post i includes a text X_i , which may help characterize its topic or other relevant aspects associated with user engagement.

Finally, each post i , described by the tuple (c, p, X_i) , elicits a number of positive and negative reactions, denoted by n_i^+ and n_i^- , respectively, from users who viewed it. Our goal is to investigate how b_c , b_p , and X_i are associated with user engagement, as measured by n_i^+ and n_i^- .

4 Methodology

This study follows a six-stage analytical process: (1) data collection from Facebook via the CrowdTangle API, (2) data preprocessing, (3) computation of political bias metrics, (4) content characterization through toxicity scoring and news topic classification, (5) computation of audience engagement metrics, and (6) statistical modeling using a Generalized Linear Mixed Model (GLMM) with a Beta distribution. Fig 1 provides an overview of the analytical framework, illustrating the relationships between publishers, news content, and the audience. The figure summarizes the key metrics defined in the following sections. In particular, it illustrates how a publisher’s estimated political bias (b_p) and the ideological leaning of the shared news article (b_c) combine to produce the Bias Discrepancy (Δ_b). These exposure dynamics, together with the thematic

classification of the news articles and the toxicity of the publisher's post text, are then related to audience engagement, captured through the Reaction Score (RS) and the Consensus Index (CI).

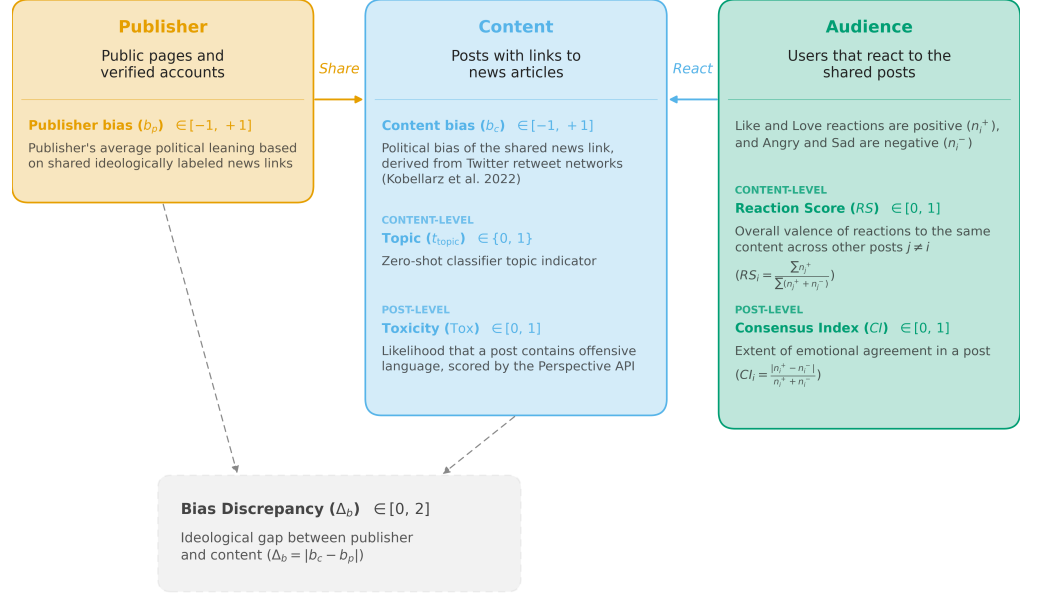


Fig 1. Conceptual framework of the study. The diagram maps the propagation of political news from publishers to the audience, delineating the core analytical variables connecting ideological exposure to audience engagement.

4.1 Data collection

The collection process relied on a dataset of 2,317 unique political news links (URLs) originally introduced by [28]. These links were identified from Twitter posts containing election-related hashtags. Each link was labeled with a relative content political bias score (originally denoted $RP(H)_{content}$ in [28], hereafter b_c), which ranges continuously from -1.0 (left) to $+1.0$ (right). Values between -0.3 and $+0.3$ are considered centrist.

The score represents the weighted average polarity of users who retweeted the content and is derived from hashtags classified through a combination of manual annotation and a semi-supervised labeling procedure [28]. Because the score is assigned to the news link itself rather than to the platform where it was shared, it provides a platform-independent proxy for the ideological orientation of the content. To assess the robustness of this metric and mitigate potential circularity issues, the original study evaluated the approach using synthetic null models, thereby providing support for using b_c as an estimator of political bias.

Using the CrowdTangle API [32], we collected public Facebook posts sharing the selected links, including metadata, textual content, and reaction counts, specifically for the four main reactions: Like, Love, Angry, and Sad. The political bias metric b_p characterizes the editorial behavior of publishers, specifically which ideologically labeled news they choose to share, rather than the reaction behavior of the audience.

4.2 Data preprocessing

A four-step data cleaning pipeline was applied to ensure data quality, as summarized in Table 1. The cleaning process consisted of three filtering steps applied sequentially.

First, duplicate entries were removed. Second, posts with inaccessible or missing textual content were excluded, including deleted posts and broken links. Third, posts with fewer than 50 reactions were removed. The minimum reaction threshold was determined through exploratory analysis of sample size stability and metric dispersion across increasing cutoff values. An empirical assessment of this threshold selection is provided in S1 Appendix.

After applying all filtering steps, the final analytical dataset comprised 4,584 posts. Using this cleaned dataset, we next define the analytical variables used to characterize political discrepancy between publishers and content, properties of the shared content, and patterns of audience engagement.

Table 1. Data cleaning pipeline and post retention at each step.

Step	Filtering procedure	Remaining (N)	Loss (%)
1	Raw posts	28,181	–
2	Duplicates removed	25,473	9.61%
3	Inaccessible text removed	17,973	29.45%
4	Posts with < 50 reactions excluded	4,584	74.49%

4.3 Political bias metrics

To operationalize ideological discrepancy between publishers and shared content, we define two complementary metrics: publisher political bias (b_p) and Bias Discrepancy (Δ_b).

We estimate each publisher’s political bias (b_p) by extending the polarization heuristic introduced by [28] from hashtag-based to content-interaction-based estimation. Specifically, b_p captures a publisher’s average political leaning based on their history of sharing ideologically labeled news links (b_c). Prior work has evaluated the validity and consistency of this methodological framework across distinct political contexts (e.g., Brazil and Canada), showing strong agreement with expert labeling and reliable identification of ideological clusters [28]. In addition, the original study assessed potential circularity issues using synthetic null models, providing further support for the robustness of the metric.

In our methodological framework, “content” refers exclusively to the shared news articles (the URLs), whose political bias scores were derived from Twitter retweet networks. It does not include the text or commentary added by the publisher alongside the link. The publisher bias score is computed as follows:

$$b_p = \frac{n_R w_R - n_L w_L}{n_L w_L + n_C w_C + n_R w_R},$$

where n_R , n_L , and n_C represent the number of interactions with right-leaning, left-leaning, and center-leaning content, respectively. The weights w_R , w_L , and w_C are class-balancing factors defined as $w_R = \frac{n_R + n_C}{N}$, $w_L = \frac{n_L + n_C}{N}$, and $w_C = \frac{n_R + n_L}{N}$, where $N = n_R + n_L + n_C$ denotes the total number of interactions. Values of b_p approaching +1.0 indicate right-leaning tendencies, whereas values approaching −1.0 indicate left-leaning tendencies.

The Bias Discrepancy (Δ_b), which measures the ideological gap between the publisher and the content they share, serves as the key operationalization of ideological discrepancy in RQ1. It is defined as the absolute difference between b_p and b_c :

$$\Delta_b = |b_c - b_p|.$$

Values close to 0.0 indicate low ideological discrepancy, while higher values indicate greater divergence between the publisher’s stance and the shared content’s ideological orientation. This metric is computed per post, with one value for each (publisher, content) pair, enabling us to quantify selective exposure and examine its relationship with audience engagement.

Having operationalized the political dimension of the publisher–content relationship, we next characterize the content itself.

4.4 Content characterization

Beyond political discrepancy, we characterize each post along two complementary dimensions derived from different sources of content. First, we analyze the linguistic tone of the publisher’s post text using toxicity scoring. Second, we identify the thematic topic of the linked news article using zero-shot classification.

Toxicity (Tox) captures the presence of hostile or inflammatory language in the text written by the publisher alongside the shared news link. We employed Google’s Perspective API [33], which returns a continuous score $\text{Tox} \in [0, 1]$ indicating the likelihood that a text may be perceived as toxic (e.g., insults, threats, or hate speech). Instead of translating the content, we processed Facebook posts directly in their original language (Portuguese). This decision follows findings by [34], who showed that automatic translation from Brazilian Portuguese to English can artificially reduce toxicity scores, particularly in highly toxic comments.

To characterize the thematic content of the shared news articles, we employed the multilingual transformer model mDeBERTa-v3-base for zero-shot topic classification. The procedure used to define topic categories and select the final classification model is described in S2 Appendix. The resulting topic labels allow us to examine how audience engagement varies across different categories of political news content.

4.5 Audience engagement metrics

With the content fully characterized, we now define the metrics that capture the audience’s collective response to each post. These metrics are computed from the reaction counts recorded on Facebook posts and serve as the primary observational signals for both research questions.

Reaction counts on Facebook provide a direct signal of collective emotional response to political content [2], but are typically analyzed at the level of individual posts, without distinguishing the affective character of the news content itself from the dynamics of a particular publisher–audience interaction. We propose the Reaction Score (RS), a metric designed to capture the overall **external** emotional valence of audience responses to the content shared in a given post. Because the same news link may be shared by multiple publishers, we compute RS_i using the aggregate reactions from all other posts sharing the same content c , excluding the focal post i to avoid circularity. Let $\mathcal{P}_c$ denote the set of all posts sharing content c . Then:

$$RS_i = \frac{\sum_{\substack{j \in \mathcal{P}_c \\ j \neq i}} n_j^+}{\sum_{\substack{j \in \mathcal{P}_c \\ j \neq i}} (n_j^+ + n_j^-)},$$

where n_j^+ and n_j^- denote the counts of positive reactions (Like, Love) and negative reactions (Angry, Sad) for post j . The score ranges from 0 to 1, where higher values indicate predominantly positive audience responses to the content. In short, RS_i varies

only slightly across posts — since it does not account for the reactions in i — and can be interpreted as the expected emotional valence of post i .

To capture the degree of agreement or conflict among audience members reacting to the focal post, we introduce the Consensus Index (CI). The absolute value ensures that consensus reflects agreement regardless of whether reactions are predominantly positive or negative:

$$CI_i = \frac{|n_i^+ - n_i^-|}{n_i^+ + n_i^-}.$$

Values close to 1 indicate strong consensus (reactions are predominantly one type), while values near 0 indicate disagreement (reactions are evenly split between positive and negative).

While the Reaction Score captures the direction of emotional valence at the content level, the Consensus Index measures the degree of agreement among users reacting to the focal post. For example, a post that receives mostly negative reactions will exhibit a low Reaction Score but a high Consensus Index, indicating strong agreement around a negative evaluation. Conversely, a post receiving similar numbers of positive and negative reactions will have a low Consensus Index, reflecting genuine disagreement. Because RS_i aggregates reactions from posts $j \neq i$ sharing the same content, while CI_i is computed from the focal post’s own reactions n_i^+ and n_i^- , the two metrics are not mathematically redundant. The correlation between them is $r \approx 0.75$, reflecting that the same content tends to be associated with similar reactions across publishers, while still leaving substantial independent variance in each measure.

4.6 Statistical modeling

To address RQ2 (what factors are most strongly associated with variation in audience consensus), we model the effects of b_p , b_c , Δ_b , RS , Tox , and topical categories on the Consensus Index (CI). Including RS allows us to test whether the emotional valence of audience reactions is associated with the degree of consensus expressed in those reactions.

The data exhibit a cross-classified hierarchical structure: a single publisher may generate multiple posts, and the same news article may be shared by multiple publishers. To account for this dependency structure, we employ a Generalized Linear Mixed Model (GLMM) with a Beta distribution and logit link. GLMMs extend standard regression models by incorporating both fixed effects (the variables of interest, such as toxicity or political bias) and random effects that capture baseline variation across observational units. The Beta distribution is appropriate for modeling continuous outcomes bounded in the unit interval $[0, 1]$, such as our Consensus Index.

Because the Beta distribution is defined on the open interval $(0, 1)$ and the empirical Consensus Index includes boundary values, we apply the standard transformation proposed by Smithson and Verkuilen [35] to map observations into the open interval, generating CI' . Additional details on this transformation are provided in S3 Appendix.

To capture baseline heterogeneity in engagement, we include random intercepts for both publishers and content. In addition, we specify random slopes for Tox , allowing the relationship between toxicity and consensus to vary across publishers and across content items. This specification relaxes the assumption that toxicity has a homogeneous effect across all contexts.

The model is estimated using the `glmmTMB` package in R, with the following specification:

$$\begin{aligned}
CI' \sim & b_c + b_p + \Delta_b + \text{Tox} + RS \\
& + t_{\text{Economy}} + t_{\text{Education}} + t_{\text{Health}} + t_{\text{Security}} \\
& + t_{\text{Culture}} + t_{\text{Religion}} + t_{\text{Disinformation}} \\
& + t_{\text{Election}} + t_{\text{Politics}} + t_{\text{Corruption}} \\
& + (\text{Tox} \mid \text{Publisher}) \\
& + (\text{Tox} \mid \text{Content}),
\end{aligned}$$

where each $t_{\text{topic}} \in [0, 1]$ represents the zero-shot classifier score assigned to that topic for the shared news link. Because a news article may be associated with multiple topics simultaneously, the ten topic scores are included as separate continuous predictors rather than as a single categorical variable.

All continuous predictors are standardized using Z-score normalization. Potential multicollinearity among fixed effects is assessed using Variance Inflation Factors (VIF), with all values below commonly used thresholds ($\text{VIF} < 5$).

Including b_p , b_c , and Δ_b in the same model allows us to distinguish the independent contributions of publisher ideology, content ideology, and their relative mismatch. Although Δ_b is derived from b_p and b_c , VIF indicates that multicollinearity remains within acceptable levels.

5 Results

This section presents the main empirical findings.

5.1 Dataset descriptive analysis

Considering the final processed dataset, Table 2 shows that both the distribution of shared news content and publishers' bias are skewed to the left. Shared news content is predominantly left-leaning (52.79%), followed by center-oriented content (31.48%), while right-leaning content represents a smaller share (15.73%). A similar pattern is observed for publishers: left-leaning publishers account for 54.54% of the total, right-leaning publishers represent 25.90%, and center-oriented publishers are the least frequent group (19.55%).

Table 2. Distribution of political orientation for content and publishers.

Category	N	Percentage
<i>Content Orientation (N = 4,584)</i>		
Left ($b_c \leq -0.3$)	2,420	52.79%
Center ($-0.3 < b_c < 0.3$)	1,443	31.48%
Right ($b_c \geq 0.3$)	721	15.73%
<i>Publisher Orientation (N = 1,243)</i>		
Left ($b_p \leq -0.3$)	678	54.54%
Center ($-0.3 < b_p < 0.3$)	243	19.55%
Right ($b_p \geq 0.3$)	322	25.90%

Fig 2 shows the distribution of the main analytical variables. Reaction Score (RS) values are strongly skewed toward 1.0, indicating that positive reactions (Like, Love) dominate user engagement. The Bias Discrepancy (Δ_b) is concentrated at low values, suggesting that publishers tend to share content aligned with their own ideological

position. Toxicity (Tox) exhibits a right-skewed distribution, with most posts showing low toxicity but a long tail of highly toxic content. Finally, Consensus Index (CI) values are concentrated near 1.0, indicating high levels of agreement in audience reactions.

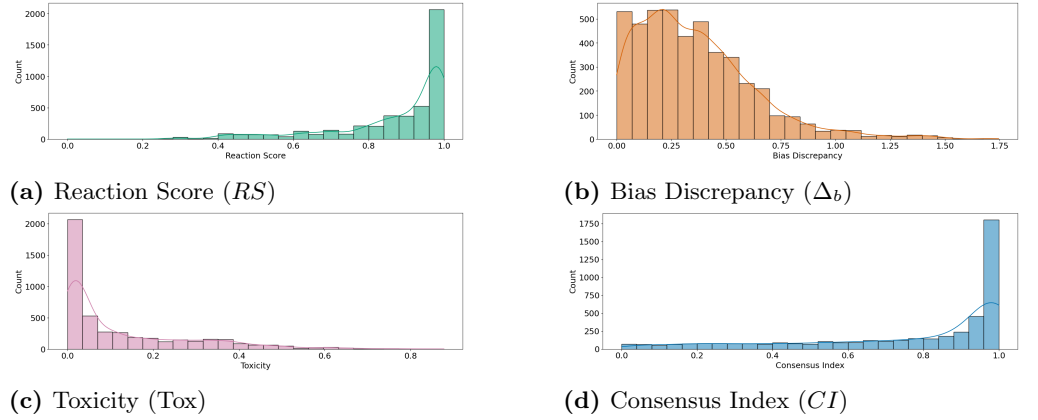


Fig 2. Distribution of the main analytical variables. Curves represent kernel density estimates.

Fig 3 illustrates the distribution of topic probabilities across the dataset. Topics related to politics and elections ($t_{Politics}$, $t_{Election}$) exhibit the highest median probabilities and the greatest variability, indicating their central role in the corpus. In contrast, topics such as education and religion ($t_{Education}$, $t_{Religion}$) show lower median probabilities, suggesting they are less prevalent overall. However, the presence of high-probability outliers indicates that these topics still appear prominently in specific posts.

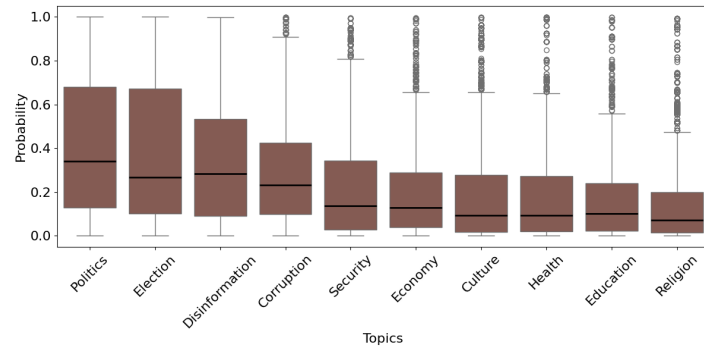


Fig 3. Distribution of probability scores across identified topics. Boxplots represent the predicted probability for each topic across posts.

5.2 Association between political bias and engagement

Fig 4 illustrates how publisher bias (b_p) and the political leaning of shared news (b_c) are associated with three engagement dimensions: Reaction Score, Consensus Index, and Toxicity. Statistical significance was assessed using Welch's ANOVA and Games-Howell post-hoc tests to account for unequal variances.

Reaction Score (Fig. 4a): Reaction Score varies systematically across ideological groups. Left-leaning publishers exhibit lower average Reaction Scores than center- and

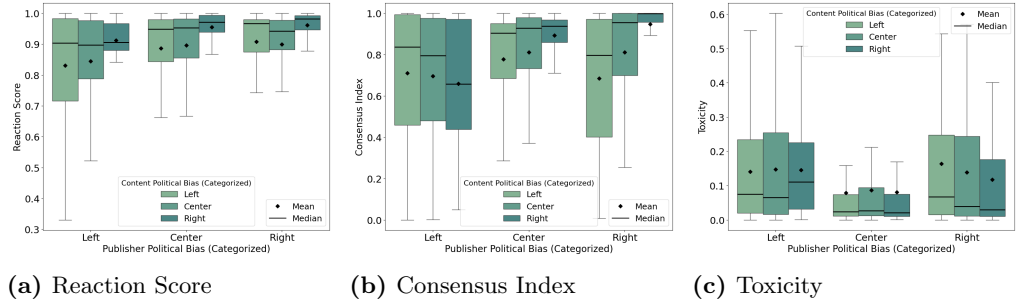


Fig 4. Distribution of engagement metrics across publisher bias and shared news political leaning.

right-leaning publishers ($p < 0.001$). Across content categories, right-leaning news is associated with higher Reaction Scores than left- and center-leaning content ($p < 0.05$). Among center-leaning publishers, right-leaning content received higher Reaction Scores than other content types ($p < 0.001$).

Consensus Index (Fig. 4b): Audience consensus also varies across ideological configurations. Posts by left-leaning publishers exhibit a lower Consensus Index than those by center- and right-leaning publishers within the same content category ($p < 0.001$). Right-leaning content is generally associated with a higher Consensus Index, whereas left-leaning content shows lower agreement across groups (both $p < 0.001$).

This pattern becomes more pronounced in the presence of an ideological discrepancy. The lowest levels of Consensus Index are observed when left-leaning publishers share right-leaning content, indicating a strong breakdown of agreement in this configuration. Right-leaning publishers sharing left-leaning content also show reduced Consensus Index relative to aligned interactions, but to a lesser extent. This asymmetry is consistent with prior work showing that cross-cutting exposure may intensify disagreement [10, 28].

Toxicity (Fig. 4c): Toxicity patterns differ from those observed for Reaction Score and Consensus Index. Center-leaning publishers exhibit lower toxicity levels than both left- and right-leaning publishers across content categories ($p < 0.001$). With respect to content orientation, we find no consistent evidence supporting an “out-group hostility” effect: toxicity levels toward ideologically opposing content are not significantly different from those toward aligned content, despite some variation in median values across groups.

Overall, these results indicate that ideological discrepancy between publishers and shared content is associated with systematic differences in audience reactions and levels of agreement. Notably, right-leaning publishers exhibit higher levels of audience consensus, consistent with prior findings on right-leaning cohesion in the Brazilian context [12, 28].

5.3 Association between bias discrepancy and engagement

To examine how ideological discrepancy between publishers and shared content relates to audience engagement, we analyze Reaction Score (RS), Consensus Index (CI), and Toxicity (Tox) across levels of Bias Discrepancy (Δ_b). For interpretability, Δ_b was discretized into quartiles, ranging from “Very Low” (high alignment) to “Very High” (strong ideological mismatch). Statistical significance was assessed using Welch’s ANOVA and Games-Howell post-hoc tests.

Reaction Score (Fig. 5a): Reaction Score is statistically lower in the “Very Low” ideological divergence category than in all other levels of ideological divergence ($p < 0.001$). This unexpected pattern suggests that highly aligned publisher–content

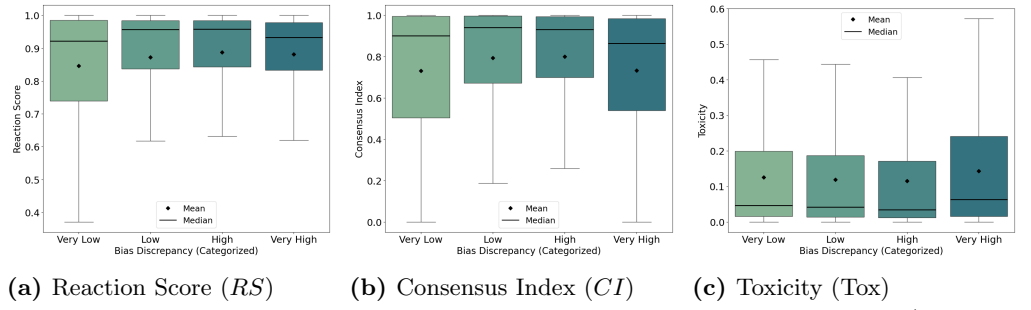


Fig 5. Distribution of engagement metrics across levels of Bias Discrepancy (Δ_b , categorized into quartiles). Colors represent the four discrepancy levels, from “Very Low” to “Very High.”

pairs are associated with less positive audience reactions than moderately discrepant configurations. One possible explanation is that highly aligned posts may involve more strongly partisan content, which could elicit more polarized responses. In this direction, prior work suggests that cross-cutting exposure can shape affective reactions in complex ways [9]. However, the mechanisms underlying this pattern cannot be directly identified from the present data, so this finding should be interpreted cautiously.

Consensus Index (Fig. 5b): Consensus Index is statistically lower in the “Very Low” and “Very High” ideological divergence categories compared to the “Low” and “High” categories. No statistically significant difference is observed between the “Very Low” and “Very High” groups.

Toxicity (Fig. 5c): Toxicity also differs across divergence levels ($p < 0.001$). No differences are observed among the lower and moderate divergence groups. The “Very High” category exhibits significantly higher toxicity than all others, indicating that increases in hostile language are concentrated at the upper end of ideological discrepancy.

Overall, the results show that engagement patterns change primarily at extreme levels of ideological discrepancy. Reaction Score is lowest when publisher-content divergence is “Very Low.” The highest level of divergence is associated with lower consensus and higher toxicity. These findings provide evidence relevant to RQ1, suggesting that polarization effects emerge most clearly under conditions of strong publisher-content discrepancy.

5.4 Modeling factors associated with consensus

Table 3 presents the estimates from the Generalized Linear Mixed Model predicting the Consensus Index, addressing RQ2. The marginal R^2 is 0.742, indicating that the fixed effects explain a substantial portion of the variance in consensus. When random effects are included, the conditional R^2 increases to 0.978, reflecting substantial between-publisher and between-content heterogeneity captured by the random effects, rather than overfitting of fixed predictors.

Among the predictors, the Reaction Score (RS) shows the strongest positive association with the Consensus Index (CI) ($\beta = 0.966, p < 0.001$), indicating that posts citing content with positive repercussion on social media tend to exhibit higher levels of agreement. Content that receives predominantly negative reactions across publishers is not associated with high consensus around negative evaluations. Instead, it corresponds to more divided reactions, consistent with prior evidence that exposure to cross-cutting content tends to increase disagreement [4, 9] and with work showing that the affective properties of content shape participatory engagement [2, 8]. The estimated associations

of the remaining predictors are therefore conditional on this content-level affective signal, reflecting variation in consensus beyond what is accounted for by the emotional valence of the news content. Ideological variables also show consistent effects. Higher values of b_p and b_c (reflecting increasingly right-leaning ideological positions) are associated with higher consensus. In contrast, Bias Discrepancy (Δ_b) is negatively associated with consensus ($\beta = -0.158, p < 0.001$), indicating that greater ideological discrepancy between publishers and content corresponds to lower agreement among users.

Toxicity (Tox) is also negatively associated with consensus ($\beta = -0.151, p < 0.001$), suggesting that higher levels of hostile language are linked to more fragmented audience responses.

With respect to topical categories, posts related to education ($t_{\text{Education}}$) ($\beta = 0.119, p = 0.004$) are associated with higher consensus, whereas health-related content (t_{Health}) ($\beta = -0.106, p = 0.026$) shows lower levels of agreement.

Table 3. Results of the Mixed Beta Regression Model. Coefficients in bold are statistically significant.

Dependent variable:		Consensus Index	
Predictors	Estimates β	Conf. Interv.	p
(Intercept)	1.645	1.575 – 1.715	< 0.001
b_c	0.095	0.032 – 0.157	0.003
b_p	0.073	0.018 – 0.128	0.009
Bias Discrepancy (Δ_b)	-0.158	-0.195 – -0.120	< 0.001
Toxicity (Tox)	-0.151	-0.199 – -0.103	< 0.001
Reaction Score (RS)	0.966	0.904 – 1.029	< 0.001
t_{Economy}	0.034	-0.045 – 0.114	0.401
$t_{\text{Education}}$	0.119	0.039 – 0.200	0.004
t_{Health}	-0.106	-0.200 – -0.013	0.026
t_{Security}	0.073	-0.028 – 0.174	0.157
t_{Culture}	-0.102	-0.211 – 0.007	0.067
t_{Religion}	-0.016	-0.097 – 0.064	0.687
$t_{\text{Disinformation}}$	-0.077	-0.170 – 0.017	0.108
t_{Election}	-0.016	-0.115 – 0.082	0.748
t_{Politics}	0.102	0.009 – 0.213	0.072
$t_{\text{Corruption}}$	0.002	-0.106 – 0.102	0.974
Random Effects			
σ^2	0.03		
τ_{00} Publisher	0.13		
τ_{00} Content	0.19		
τ_{11} Publisher $\times$ Tox	0.02		
τ_{11} Content $\times$ Tox	0.03		
ρ_{01} Publisher	0.21		
ρ_{01} Content	0.06		
ICC	0.91		
N Publisher	1243		
N Content	761		
Observations	4584		
Marginal R ² / Cond. R ²	0.742 / 0.978		

The random effects analysis provides further insight into heterogeneity across communities. Fig 6 highlights publishers with the most extreme deviations in baseline consensus (random intercepts). Publishers with below-average baseline consensus are predominantly major news outlets (e.g., “UOL Notícias,” a popular news outlet in Brazil), suggesting higher levels of disagreement within their audiences. These outlets often function as *bubble reachers* (or brokers), capable of distributing content to users across diverse ideological clusters [28]. However, reaching diverse audiences is not necessarily associated with consensus. Instead, in the Brazilian context, such neutral brokers are associated with higher levels of incivility and disagreement compared to partisan accounts [10]. This pattern is consistent with prior work on the hostile media effect [36], where partisan audiences react critically to perceived bias in mainstream news coverage, and discussions tend to be more fragmented and contentious. In contrast, publishers with the above-average baseline consensus tend to be more niche or

highly partisan pages (e.g., “Jair M Bolsonaro,” representing the account of the far-right candidate for president of Brazil in 2018), which exhibit more homogeneous audience responses.

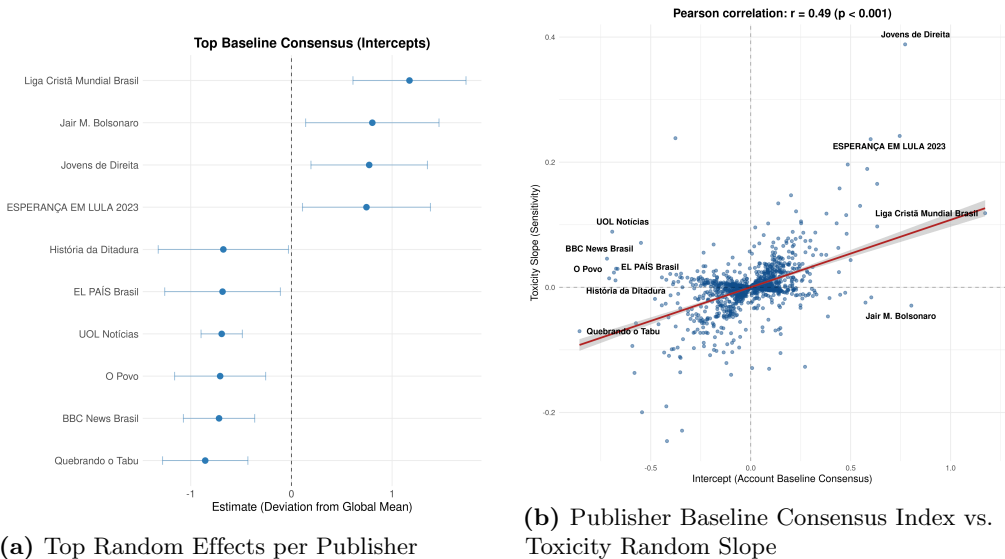


Fig 6. Analysis of publisher-level heterogeneity based on Mixed Beta GLM Random Effects.

Extending this analysis, Fig 6b shows a strong positive correlation between publishers’ baseline Consensus Index and their sensitivity to toxicity ($r = 0.49, p < 0.001$). This indicates that publishers with higher baseline agreement also tend to exhibit more positive toxicity slopes relative to the global average. This pattern suggests that, in highly cohesive communities, toxicity is not necessarily associated with internal fragmentation, but rather co-occurs with higher agreement. One possible interpretation, consistent with Social Identity Theory [37], is that hostile language in strongly partisan contexts may reflect shared identity. In this framework, incivility can be understood as a signal of in-group alignment [38], where the derogation of out-groups through hostile rhetoric may be associated with stronger internal cohesion and heightened core audience engagement [39]. This pattern is particularly relevant in the Brazilian context, where polarized digital spaces appear to be associated with high-engagement content linked to signals of ideological purity and collective commitment.

6 Conclusion

This study examines how ideological discrepancy between publishers and news content is associated with audience engagement on Facebook during Brazil’s 2018 presidential election. Across three engagement dimensions, we find evidence of nonlinear and asymmetric patterns. Audience consensus is reduced at both extremes of the discrepancy spectrum — under very high ideological mismatch and, unexpectedly, also under very high alignment. Reaction Score is lowest when publishers share content closely matching their own ideological position, an unexpected pattern that may reflect characteristics of highly aligned partisan content, although the mechanisms underlying this association remain unclear. Toxicity, in contrast, increases primarily at the upper end of ideological discrepancy. We further observe an asymmetry in our dataset:

right-leaning publishers sharing right-leaning content are associated with higher consensus and more positive reactions, whereas left-leaning publishers tend to exhibit lower consensus. These findings are consistent with prior work documenting stronger internal cohesion within right-leaning communities in the Brazilian context [12, 28], although they should be interpreted in light of the sample composition and the relative underrepresentation of right-leaning publishers compared to left-leaning publishers in the dataset. Nevertheless, the consistency of these patterns across multiple analytical approaches supports their robustness.

Regarding affective polarization, our results suggest that hostility does not exhibit a linear pattern with ideological discrepancy. Instead, elevated toxicity and reduced consensus tend to be concentrated at extreme levels of divergence. At the same time, we observe heterogeneity across communities: although toxicity is negatively associated with consensus overall, this relationship reverses in some highly cohesive partisan communities, where higher toxicity co-occurs with higher agreement. This pattern suggests that, in certain contexts, hostile language may be associated with in-group alignment rather than internal disagreement, a possibility consistent with theories of social identity [37–39].

These results should be interpreted in light of several limitations. First, ideological labels for news content were derived from Twitter-based measures [28], while engagement outcomes were observed on Facebook. Although this cross-platform approach enables large-scale analysis without relying on self-reported affiliations, the observed patterns remain shaped by Facebook’s specific affordances. In particular, the platform’s emphasis on positive reactions might inflate both the Reaction Score and the Consensus Index, leading our results to primarily capture relative differences between groups rather than absolute levels of sentiment.

Second, right-leaning content and publishers are underrepresented in the dataset. While this reflects the underlying data distribution, it requires caution when generalizing findings for this group. Third, although the Reaction Score is computed using reactions from other posts to reduce tautological dependence with the Consensus Index, some residual interdependence between these measures might persist. Fourth, the temporal gap between the original Twitter data and Facebook data collection may introduce survivorship bias, as only posts that remained accessible were analyzed. In addition, CrowdTangle data is limited to public pages and groups, excluding private interactions.

Finally, toxicity was measured using the Perspective API, a widely adopted tool but with known limitations, particularly when applied to Portuguese-language content. Prior work documents lower classification performance for Portuguese compared to English [40], as well as limited sensitivity to conversational context, sarcasm, and culturally specific forms of political expression [41, 42]. Processing content directly in Portuguese helps mitigate translation-related distortions [34, 43], but toxicity scores should nonetheless be interpreted as approximate indicators of hostile language rather than exhaustive measures of political incivility [44]. However, as a widely used solution, Perspective API provides a consistent baseline, and the persistence of statistically significant toxicity patterns despite this inherent measurement noise supports the observed relationship between incivility and ideological discrepancy.

Future research could extend this work by incorporating more context-aware and culturally grounded models of toxicity, as well as by examining whether similar patterns emerge in other political contexts or platforms with different interaction affordances.

Ethical considerations

All data used in this study were collected from publicly accessible Facebook posts via the now-deprecated CrowdTangle API. Data collection was conducted prior to the tool’s

discontinuation, in full compliance with the platform’s Terms of Service and data use policies active at the time of access. The dataset comprises content exclusively from public pages (e.g., media outlets, political figures, and organizations) operating in the public sphere. Therefore, while we report actor-specific heterogeneity to demonstrate how distinct editorial lines or political stances interact with consensus, this analysis does not expose private user data. No personally identifiable information from private citizens was analyzed individually. Toxicity scores were computed exclusively based on the textual content of the posts published by these publishers, while metrics involving the audience (such as the Consensus Index, detailed in Section 4.5) are reported solely at the aggregate level.

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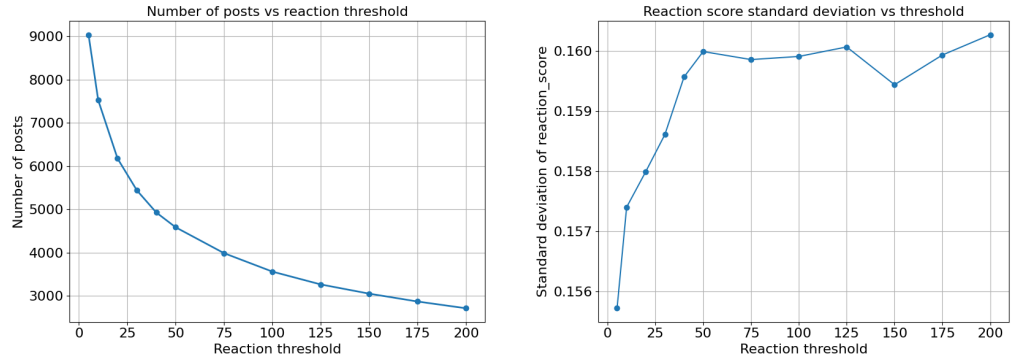
Supporting Information

S1 Appendix

Empirical assessment of the minimum reaction threshold

This appendix presents the empirical analysis used to determine the minimum reaction threshold applied in the preprocessing stage.

As shown in Fig 7a, the number of available posts decreases sharply as the minimum reaction threshold increases but stabilizes progressively beyond 50 reactions. More importantly, Fig 7b shows that the standard deviation of the Reaction Score increases substantially at very low reaction counts and stabilizes after approximately 50 reactions, indicating greater stability in the variance of the Reaction Score. These patterns suggest that engagement metrics computed from posts with very few reactions are more sensitive to noise.



(a) Number of available posts as a function of the minimum reaction threshold. (b) Standard deviation of the Reaction Score across thresholds.

Fig 7. Empirical assessment of the minimum reaction threshold. Panel (a) shows the reduction in the number of available posts as the minimum reaction cutoff increases. Panel (b) shows the standard deviation of the Reaction Score across thresholds, illustrating that dispersion increases substantially at very low reaction counts but stabilizes beyond approximately 50 reactions. This stabilization suggests that metrics computed from posts with fewer reactions are more sensitive to noise, supporting the use of 50 reactions as the minimum threshold.

S2 Appendix

Topic definition and model selection

To characterize the thematic scope of the shared news content, we first conducted an exploratory Latent Dirichlet Allocation (LDA) analysis [45] on the textual content of the linked news articles. The resulting latent themes informed the definition of ten topical categories used in the analysis: economy, education, health, security, culture, religion, disinformation, election, politics, and corruption. These predefined categories were then assigned to individual news articles using a zero-shot language model.

To select an appropriate zero-shot classification model, we conducted a comparative experiment with three candidate approaches. The first was the multilingual mDeBERTa-v3-base model fine-tuned on NLI data [46] (MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7)¹. The second was the BGE-M3 model [47] (MoritzLaurer/bge-m3-zeroshot-v2.0)². The third followed a translate-test pipeline using BART-Large-MNLI [48] (facebook/bart-large-mnli)³, where texts were first translated from Portuguese to English using the unicamp-dl/translation-pt-en-t5 model [49].

For evaluation, we manually labeled a random sample of 50 news links, indicating the presence of any of the 10 predefined topics (this labeled dataset will be released in a public repository after publication). Each model produced probability scores p_k for topic k . Instead of applying a single global threshold, we optimized the classification threshold independently for each topic to account for differences in model sensitivity across domains. Binary predictions were defined as

$$t_k = \begin{cases} 1, & \text{if } p_k \geq \tau_k \\ 0, & \text{otherwise.} \end{cases}$$

¹<https://huggingface.co/MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7>

²<https://huggingface.co/MoritzLaurer/bge-m3-zeroshot-v2.0>

³<https://huggingface.co/facebook/bart-large-mnli>

For each topic k , the optimal threshold τ_k was selected by maximizing the F1-score on the validation set. Specifically, we evaluated the F1-score at all unique probability values produced by the model, ensuring selection of the threshold that maximized performance while avoiding discretization effects associated with fixed-step grid searches.

Model performance was summarized using the Weighted Average F1-score across topics. This aggregation accounts for class imbalance by weighting topic-level F1-scores according to their support (number of positive instances) in the validation set.

As summarized in Table 4, the multilingual mDeBERTa-v3-base model achieved the best performance, with a Weighted F1-score of 0.825, compared to 0.780 for the translation-based BART-Large-MNLI pipeline and 0.768 for BGE-M3. Based on this evaluation, we selected mDeBERTa-v3-base for topic classification in the main analysis.

Table 4. Comparison of zero-shot classification models ($N = 50$).

Model Strategy	Approach	W-F1 Score
mDeBERTa-v3-base	Multilingual	0.825
BART-Large-MNLI	Trans. (PT→EN)	0.780
BGE-M3	Multilingual	0.768

S3 Appendix

Choice of the beta regression framework

The Consensus Index (CI) is a continuous variable bounded in the unit interval $[0, 1]$. Applying Ordinary Least Squares (OLS) regression to such data is problematic, as it violates key assumptions of homoscedasticity and normally distributed residuals. In bounded outcomes, variance is inherently mean-dependent and decreases as observations approach the boundaries of the support.

To address these issues, we employ a Beta regression framework within a Generalized Linear Mixed Model (GLMM). The Beta distribution is flexible and well-suited for modeling proportions and rates, allowing the conditional variance to vary as a function of the mean while ensuring that predicted values remain within the logical bounds of the dependent variable.

Boundary adjustment for the consensus index

A limitation of the Beta distribution is that it is defined on the open interval $(0, 1)$. However, the empirical Consensus Index includes observations with values exactly equal to 0 or 1. Because the likelihood function of the Beta distribution is undefined at these boundaries, a transformation is required prior to estimation.

Following Smithson and Verkuilen [35], we apply a symmetric transformation that compresses the data into the open interval:

$$CI' = \frac{CI(N - 1) + 0.5}{N},$$

where CI is the original Consensus Index and N is the total sample size. This transformation preserves the relative ordering and distributional shape of the data while ensuring numerical stability during maximum likelihood estimation.

Random effects structure

The dataset exhibits a hierarchical, cross-classified structure: individual publishers generate multiple posts, and the same news articles (content) are frequently shared

across different publishers. Ignoring this non-independence would lead to underestimated standard errors and inflated Type I error rates.

To account for this structure, we include random intercepts for both publishers and content. In addition, we specify random slopes for Tox at both levels. This allows the association between Tox and consensus to vary across publishers and content, relaxing the assumption of a uniform effect and capturing heterogeneity in how hostile discourse relates to consensus across different communities and publishers.